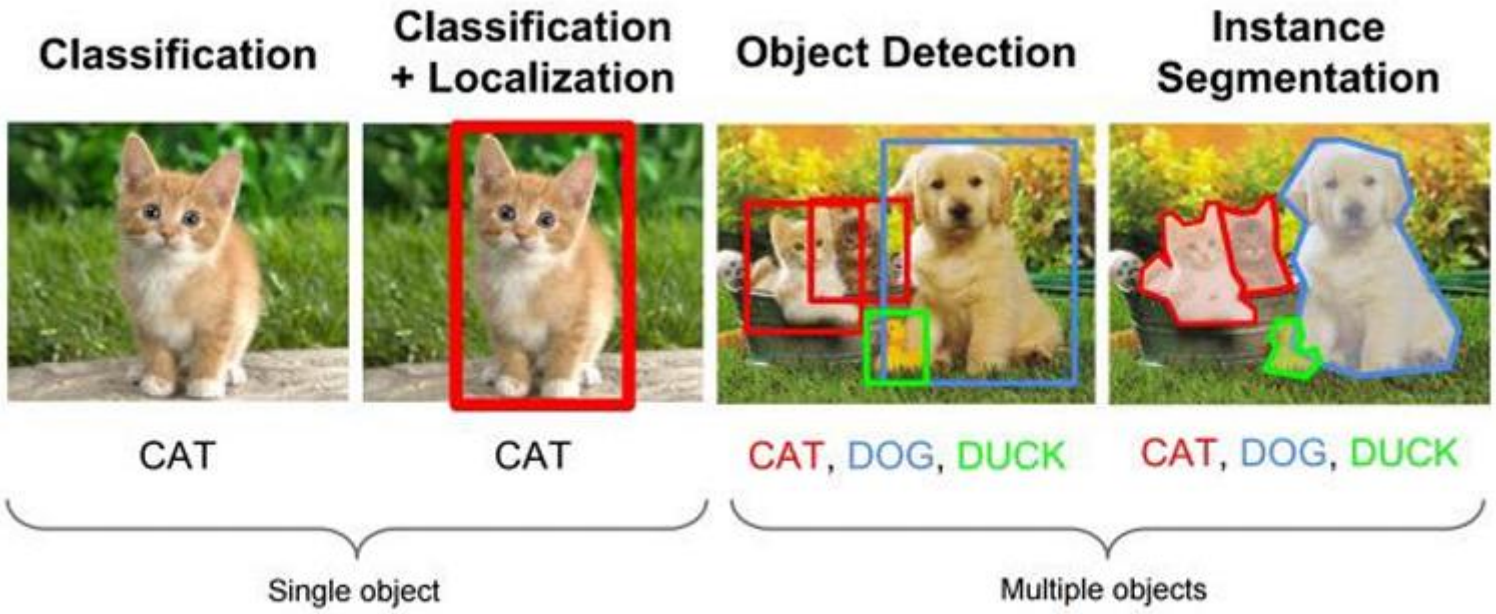
2. What is Convolutional Neural Network

Contents

2. What is Convolutional Neural Network (CNN)
   1) Convolution
   2) Image Classification Pipeline
   3) Characteristics of CNN
      - Locality
      - Invariance
      - Equivariance

Computer Vision Tasks

Image Classification
Object Detection
Semantic Segmentation
etc



2-1. Convolution

Image Data

28 × 28
= 784 pixels

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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0	0	0	0	0	0	0	0	5	25	54	25	96	140	215	215	251	254	254	254	254	254	163	0	0	0
0	0	0	0	0	0	0	0	157	254	254	254	0	254	254	254	254	254	254	254	254	241	100	0	0	0
0	0	0	0	0	0	0	0	156	117	184	214	214	156	117	49	19	19	141	254	241	158	23	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	16	173	254	246	70	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	90	247	254	219	58	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	43	203	254	250	121	15	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	43	145	254	254	229	111	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	4	137	245	254	254	238	40	0	0	0	0	0	0	0	0	0	0
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0	0	0	0	0	0	0	0	0	55	196	196	114	194	223	254	254	216	23	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	75	245	253	55	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	150	254	134	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	14	200	254	134	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	246	253	59	0	0	0	0	0	0	0
0	0	0	17	11	18	0	0	0	0	0	0	0	0	67	254	254	170	0	0	0	0	0	0	0	0
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0	0	0	29	203	254	254	254	215	254	254	254	250	95	11	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	5	80	155	155	156	111	58	58	36	0	0	0	0	0	0	0	0	0	0	0	0	0
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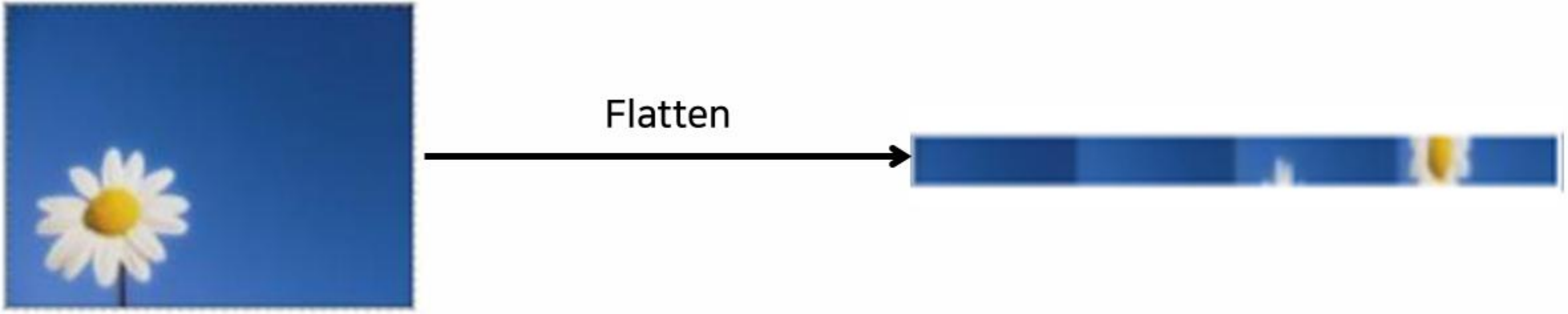
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

2-1. Convolution



2-1. Convolution

Fully connected models treat images as flat vectors, ignoring their spatial structure.

= **All pixels are treated as equally important.**

28 × 28
= 784 pixels

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	55	87	157	156	187	215	81	0	0	0	0
0	0	0	0	0	0	0	0	5	25	54	25	96	140	215	215	251	254	254	254	254	254	163	0	0	0	0
0	0	0	0	0	0	0	0	157	254	254	254	0	254	254	254	254	254	254	254	254	241	100	0	0	0	0
0	0	0	0	0	0	0	0	156	117	184	214	214	156	117	49	19	19	141	254	241	158	23	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	16	173	254	246	70	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	90	247	254	219	58	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	43	203	254	250	121	15	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	43	145	254	254	229	111	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	4	137	245	254	254	238	40	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	130	254	254	254	254	254	235	106	23	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	55	196	196	114	194	223	254	254	216	23	0	0	0	0	0	0	0	0
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0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	246	253	59	0	0	0	0	0	0	0	0
0	0	0	17	11	18	0	0	0	0	0	0	0	0	67	254	254	170	0	0	0	0	0	0	0	0	0
0	0	0	128	254	159	0	0	0	0	0	0	20	140	248	254	254	9	0	0	0	0	0	0	0	0	0
0	0	0	154	254	229	80	79	79	161	176	218	254	254	254	104	0	0	0	0	0	0	0	0	0	0	0
0	0	0	29	203	254	254	254	215	254	254	254	250	95	11	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	5	80	155	155	156	111	58	58	36	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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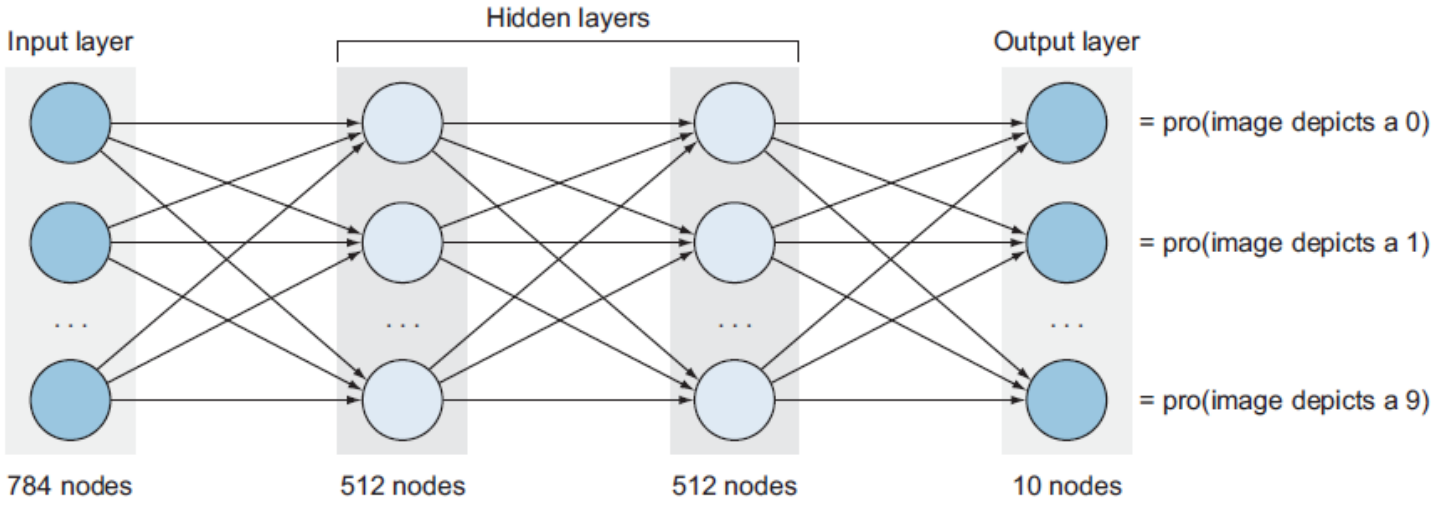


2-1. Convolution

Convolution

Use Kernel with Weight shared + Sliding Window

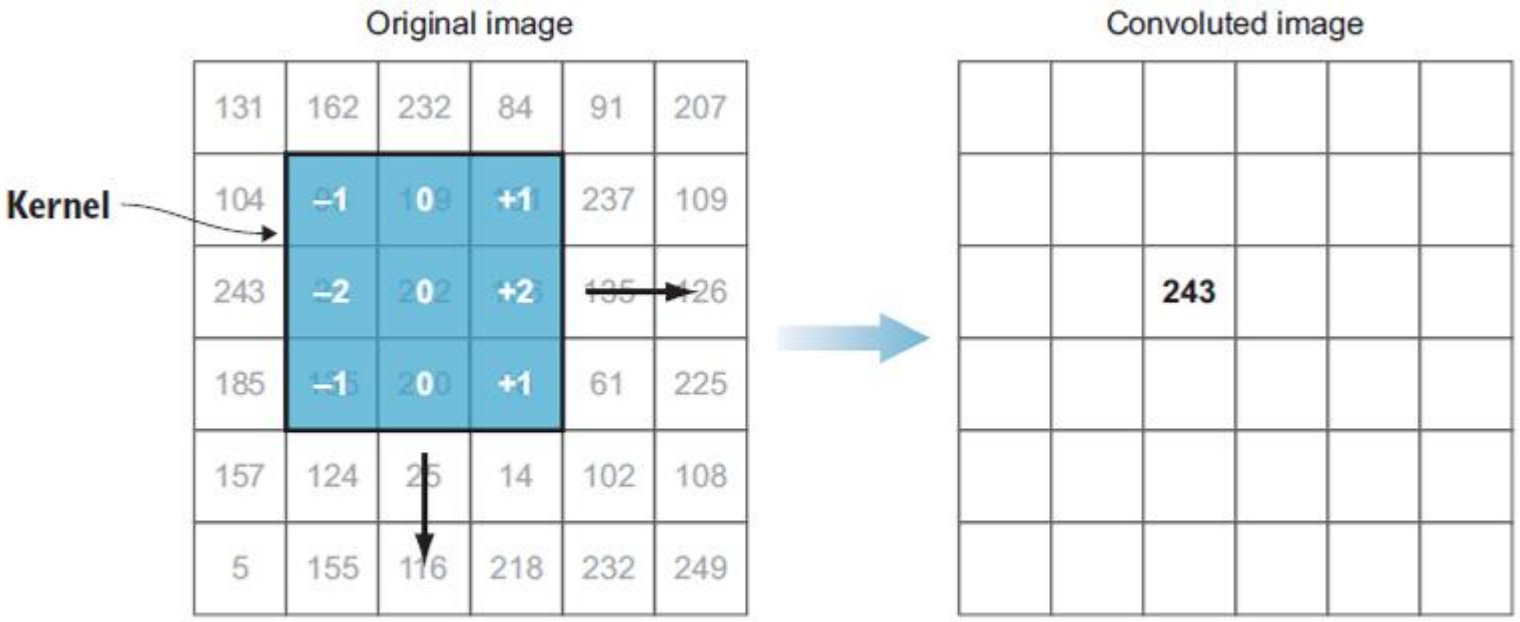


Figure 3.15 Multiplying each pixel in the receptive field by the corresponding pixel in the convolution filter and summing them gives the value of the center pixel in the new image.

2-1. Convolution

Example of Convolution

K (3x3 filter)

K_{11}	K_{12}	K_{13}
K_{21}	K_{22}	K_{23}
K_{31}	K_{32}	K_{33}

*

I (7x7 image)

I_{11} K_{11}	I_{12} K_{12}	I_{13} K_{13}	I_{14}	I_{15}	I_{16}	I_{17}
I_{21} K_{21}	I_{22} K_{22}	I_{23} K_{23}	I_{24}	I_{25}	I_{26}	I_{27}
I_{31} K_{31}	I_{32} K_{32}	I_{33} K_{33}	I_{34}	I_{35}	I_{36}	I_{37}
I_{41}	I_{42}	I_{43}	I_{44}	I_{45}	I_{46}	I_{47}
I_{51}	I_{52}	I_{53}	I_{54}	I_{55}	I_{56}	I_{57}
I_{61}	I_{62}	I_{63}	I_{64}	I_{65}	I_{66}	I_{67}
I_{71}	I_{72}	I_{73}	I_{74}	I_{75}	I_{76}	I_{77}

=

Output (5x5)

O_{11}	O_{12}	O_{13}	O_{14}	O_{15}
O_{21}	O_{22}	O_{23}	O_{24}	O_{25}
O_{31}	O_{32}	O_{33}	O_{34}	O_{35}
O_{41}	O_{42}	O_{43}	O_{44}	O_{45}
O_{51}	O_{52}	O_{53}	O_{54}	O_{55}

$$O_{11} = I_{11}K_{11}+I_{12}K_{12}+I_{13}K_{13}+I_{21}K_{21}+I_{22}K_{22}+I_{23}K_{23}+I_{31}K_{31}+I_{32}K_{32}+I_{33}K_{33}+bias$$

2-1. Convolution

Example of Convolution

K (3x3 filter)

K_{11}	K_{12}	K_{13}
K_{21}	K_{22}	K_{23}
K_{31}	K_{32}	K_{33}

$*$

I (7x7 image)

I_{11}	K_{11} I_{12}	K_{12} I_{13}	K_{13} I_{14}	I_{15}	I_{16}	I_{17}
I_{21}	K_{21} I_{22}	K_{22} I_{23}	K_{23} I_{24}	I_{25}	I_{26}	I_{27}
I_{31}	K_{31} I_{32}	K_{32} I_{33}	K_{33} I_{34}	I_{35}	I_{36}	I_{37}
I_{41}	I_{42}	I_{43}	I_{44}	I_{45}	I_{46}	I_{47}
I_{51}	I_{52}	I_{53}	I_{54}	I_{55}	I_{56}	I_{57}
I_{61}	I_{62}	I_{63}	I_{64}	I_{65}	I_{66}	I_{67}
I_{71}	I_{72}	I_{73}	I_{74}	I_{75}	I_{76}	I_{77}

$=$

Output (5x5)

O_{11}	O_{12}	O_{13}	O_{14}	O_{15}
O_{21}	O_{22}	O_{23}	O_{24}	O_{25}
O_{31}	O_{32}	O_{33}	O_{34}	O_{35}
O_{41}	O_{42}	O_{43}	O_{44}	O_{45}
O_{51}	O_{52}	O_{53}	O_{54}	O_{55}

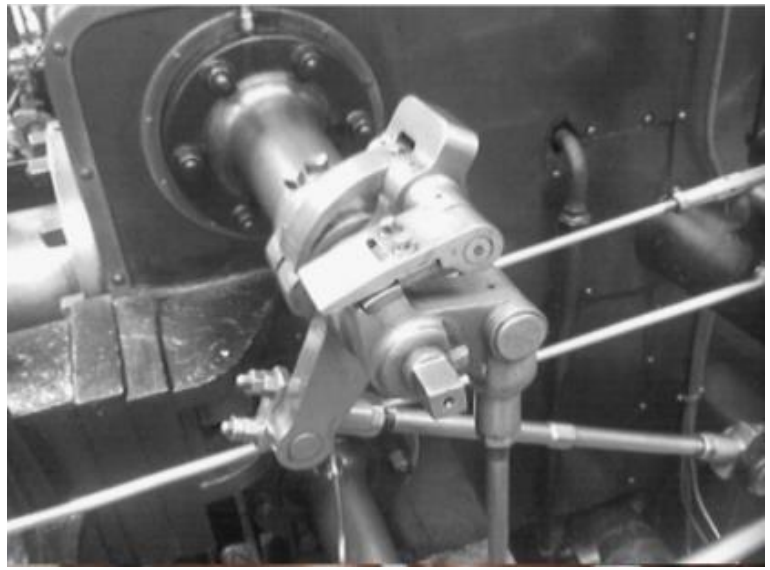
$$O_{11} = I_{11}K_{11}+I_{12}K_{12}+I_{13}K_{13}+I_{21}K_{21}+I_{22}K_{22}+I_{23}K_{23}+I_{31}K_{31}+I_{32}K_{32}+I_{33}K_{33}+bias$$

$$O_{12} = I_{12}K_{11}+I_{13}K_{12}+I_{14}K_{13}+I_{22}K_{21}+I_{23}K_{22}+I_{24}K_{23}+I_{32}K_{31}+I_{33}K_{32}+I_{34}K_{33}+bias$$

2-1. Convolution

Convolution = Filtering

because Kernel detects a specific **local pattern** in an image.

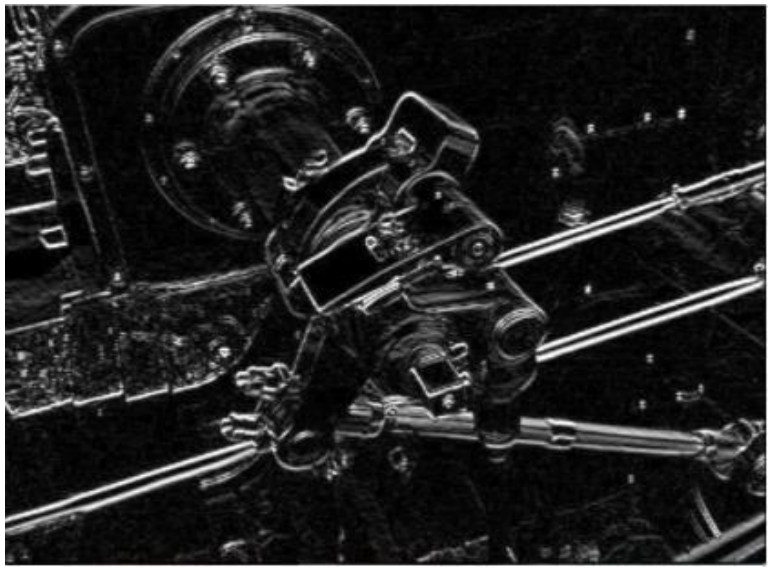


-1	0	+1
-2	0	+2
-1	0	+1

x filter

+1	+2	+1
0	0	0
-1	-2	-1

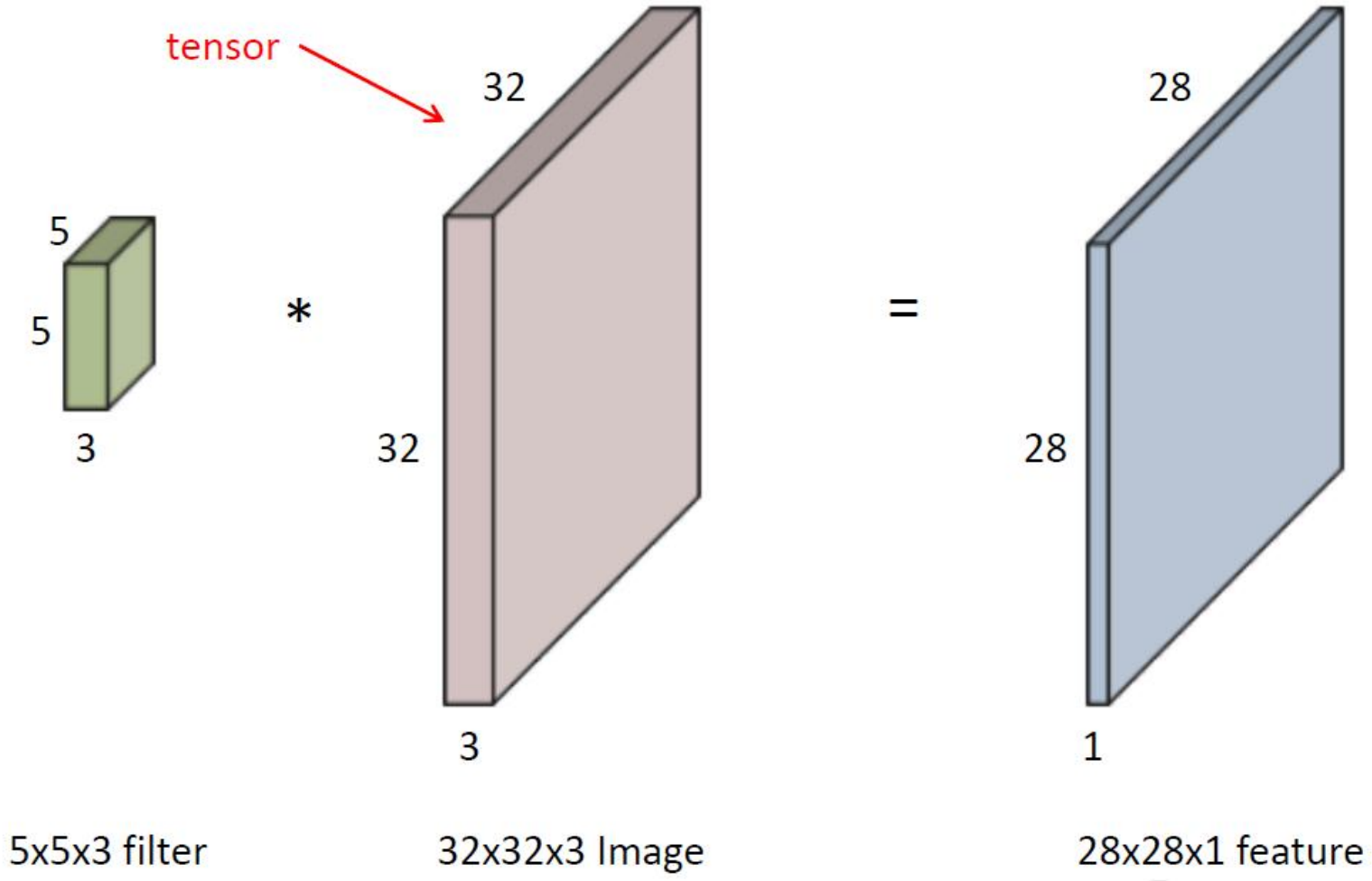
y filter



Sobel filter is used in image processing and computer vision,
particularly within **<u>edge detection</u>**

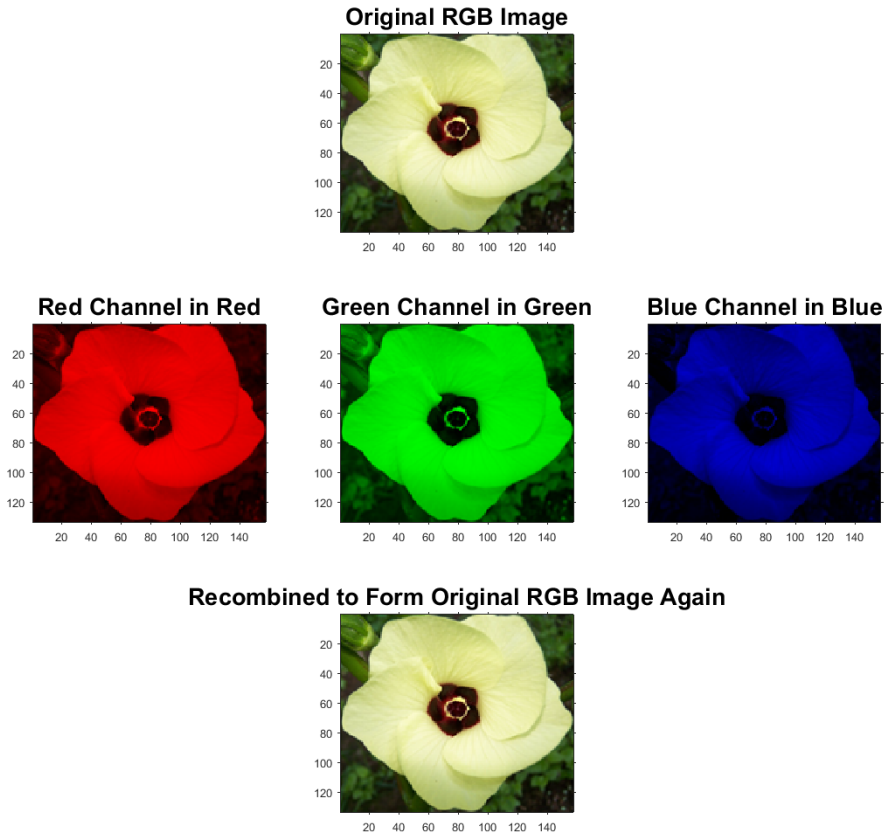
2-1. Convolution

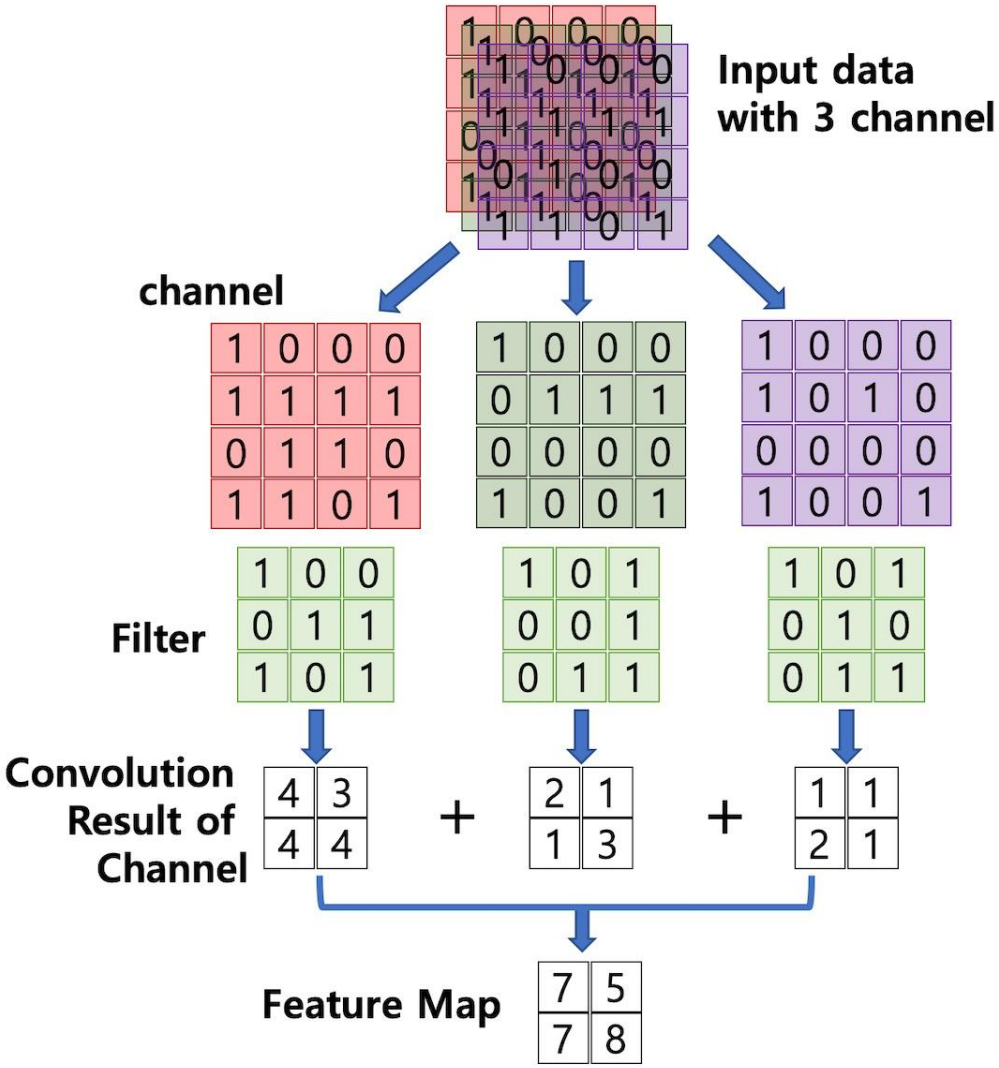
Example of Convolution



2-1. Convolution

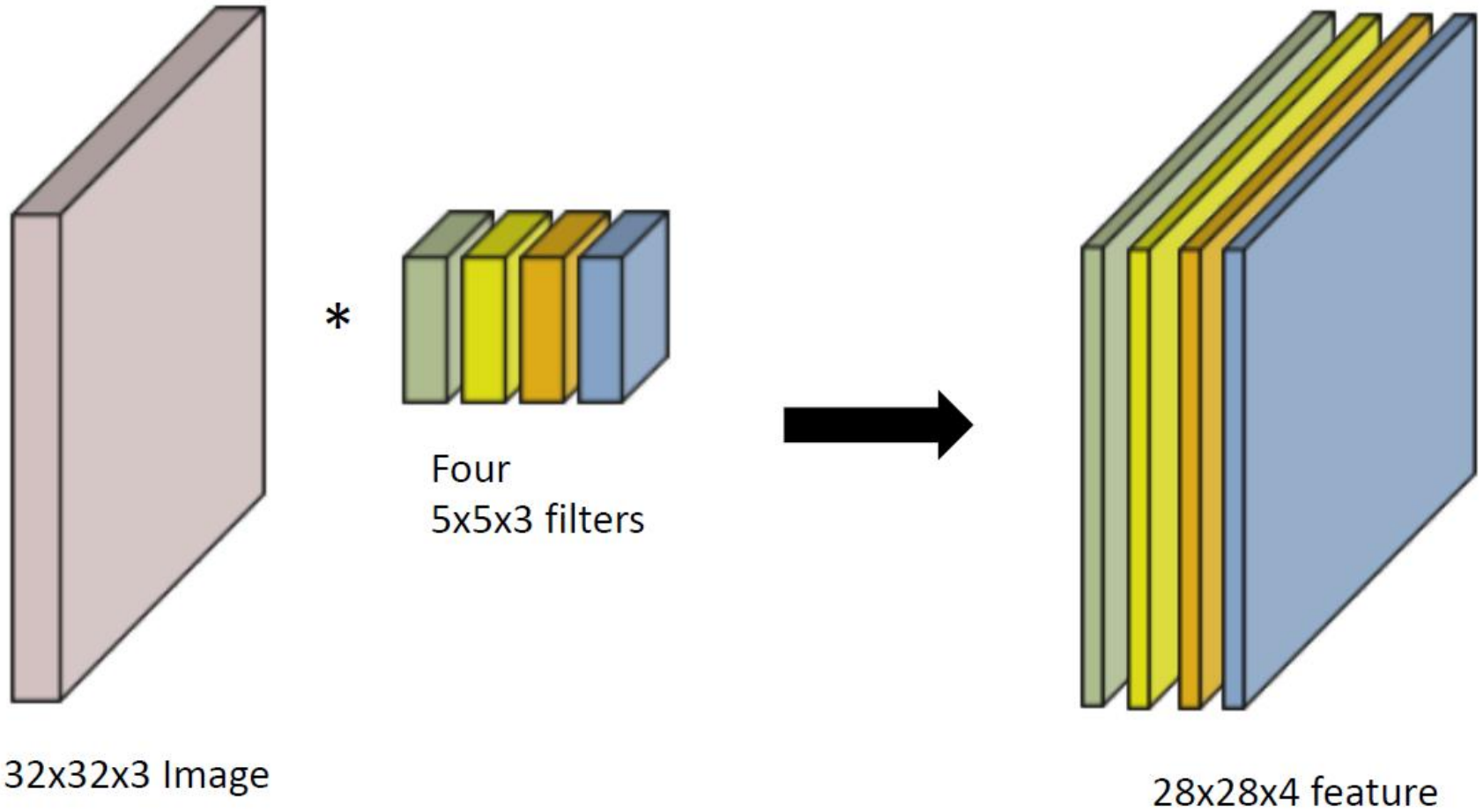
Channel





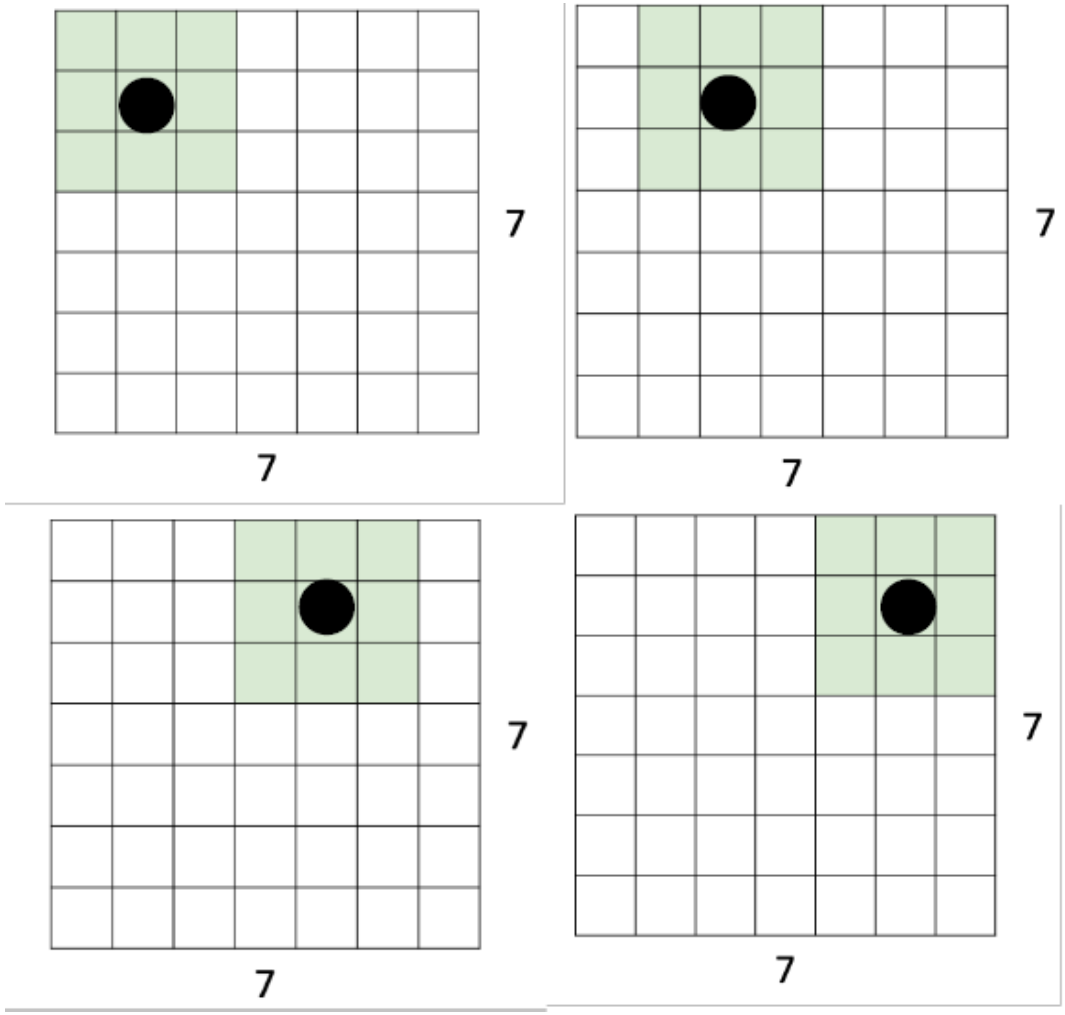
2-1. Convolution

Channel



2-1. Convolution

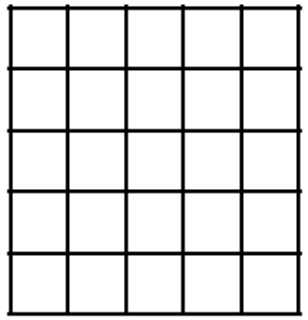
Stride



Input: 7x7

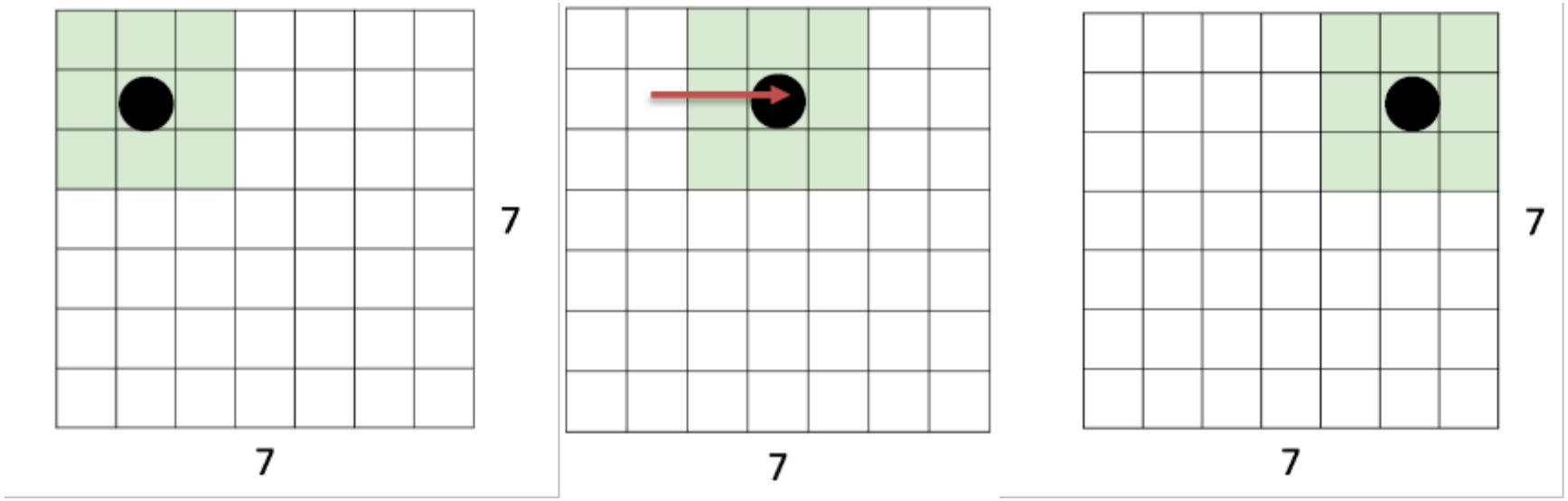
Filter: 3x3

Output: 5x5



2-1. Convolution

Stride



Input: 7x7
Filter: 3x3
Stride: 2

2-1. Convolution

Padding

0	0	0	0	0	0	0	0	0
0								0
0								0
0								0
0								0
0								0
0								0
0								0
0	0	0	0	0	0	0	0	0

Input: 7x7
Filter: 3x3
Output: 5x5

In general:
Input: W
Filter: K
Padding: P
Output: W – K + 1 + 2P

Very common: “same padding”
Set P = (K – 1) / 2
Then output size = input size

2-1. Convolution

Pooling Layer – Max

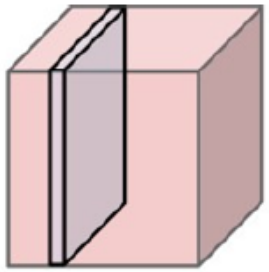
Single depth slice

x

2	3	2	4
5	**6**	6	**8**
3	2	1	0
1	2	3	**4**

y

64 x 224 x 224



Max pooling with 2x2 kernel size and stride 2

6	8
3	4

Introduces **invariance** to small spatial shifts.
No learnable parameters!

2-1. Convolution

Pooling Layer - Average

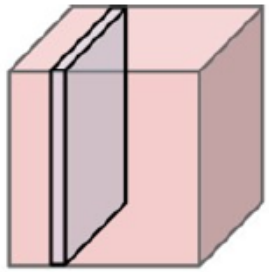
Single depth slice

x

2	3	2	4
5	6	6	8
3	2	1	0
1	2	3	4

y

64 x 224 x 224



Average pooling with 2x2 kernel size and stride 2

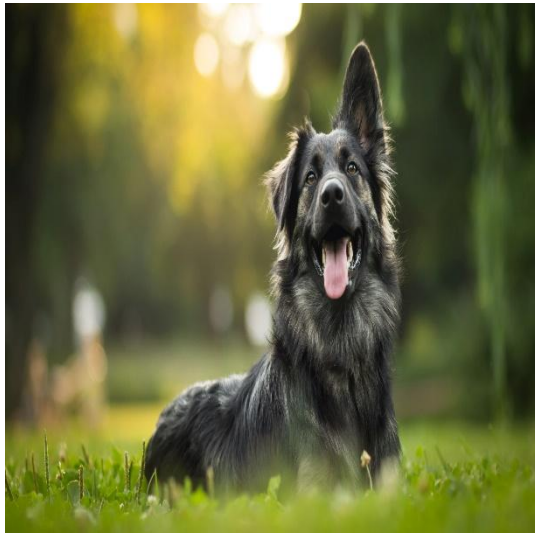
4	5
2	2

Introduces **invariance** to small spatial shifts.
No learnable parameters!

2-2. Image Classification Pipeline

Image Resize

1280 x 720

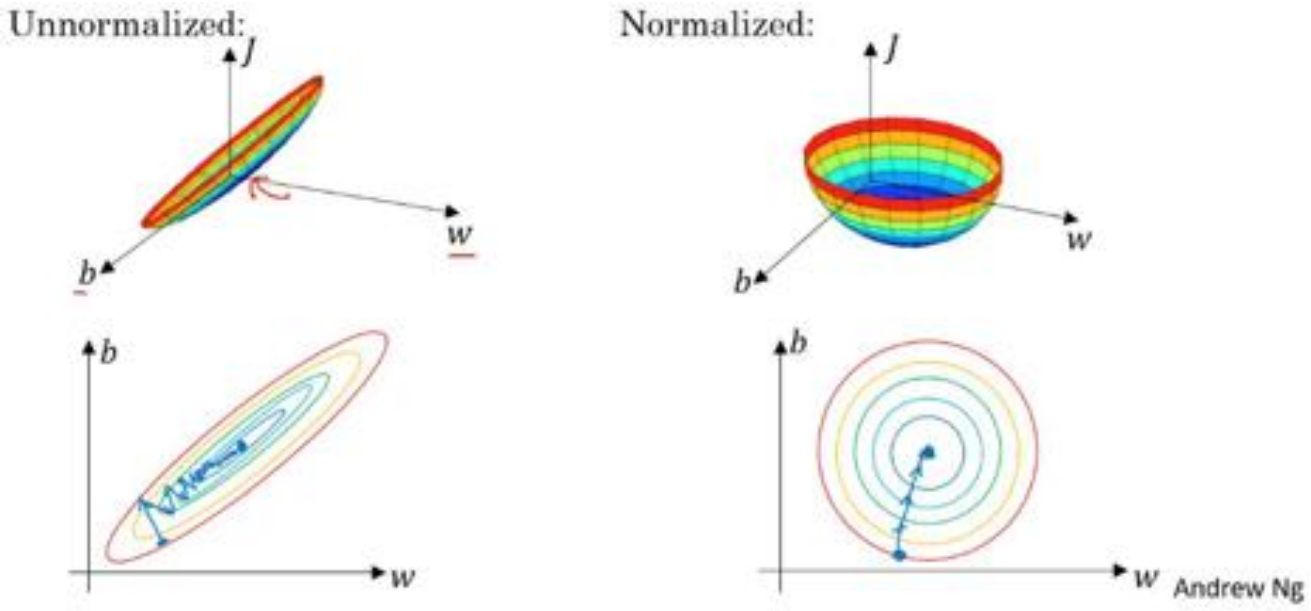


224 x 224

Image resolution is resized considering computing power

2-2. Image Classification Pipeline

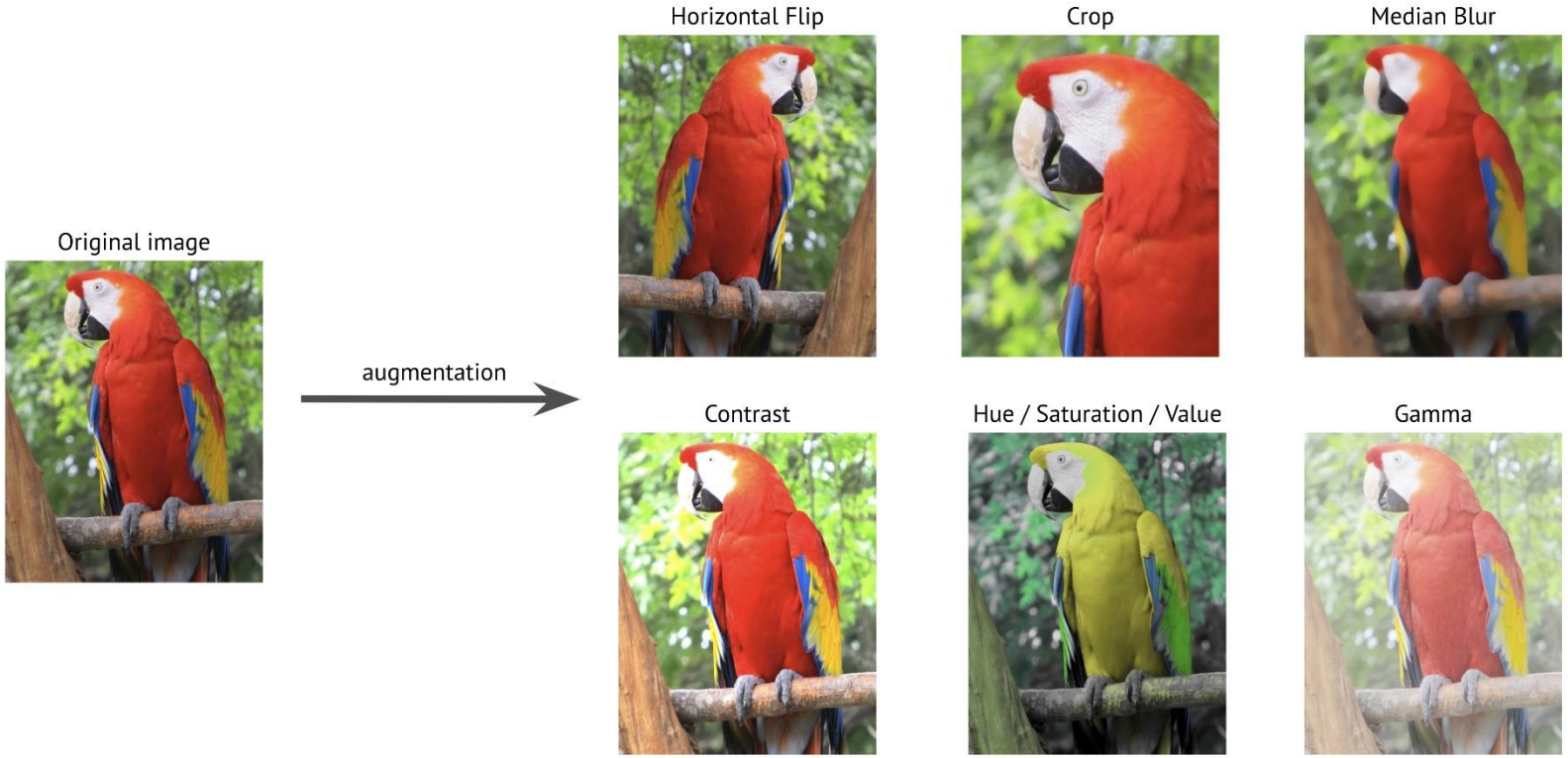
Image Normalization



In image processing, normalization is a process that changes the range of pixel intensity values

2-2. Image Classification Pipeline

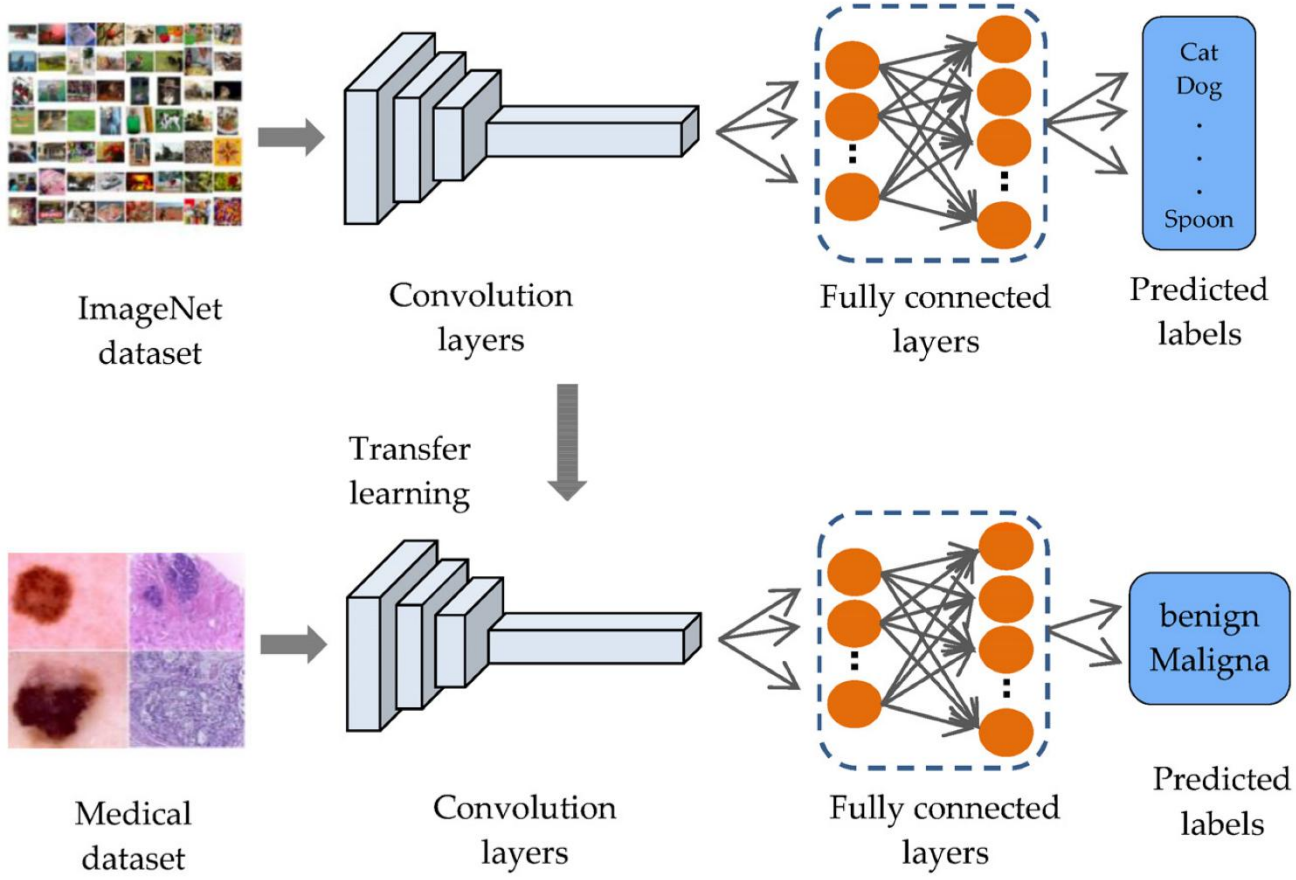
Image Augmentation



Deep neural networks require a lot of training data to obtain good results and prevent overfitting

2-2. Image Classification Pipeline

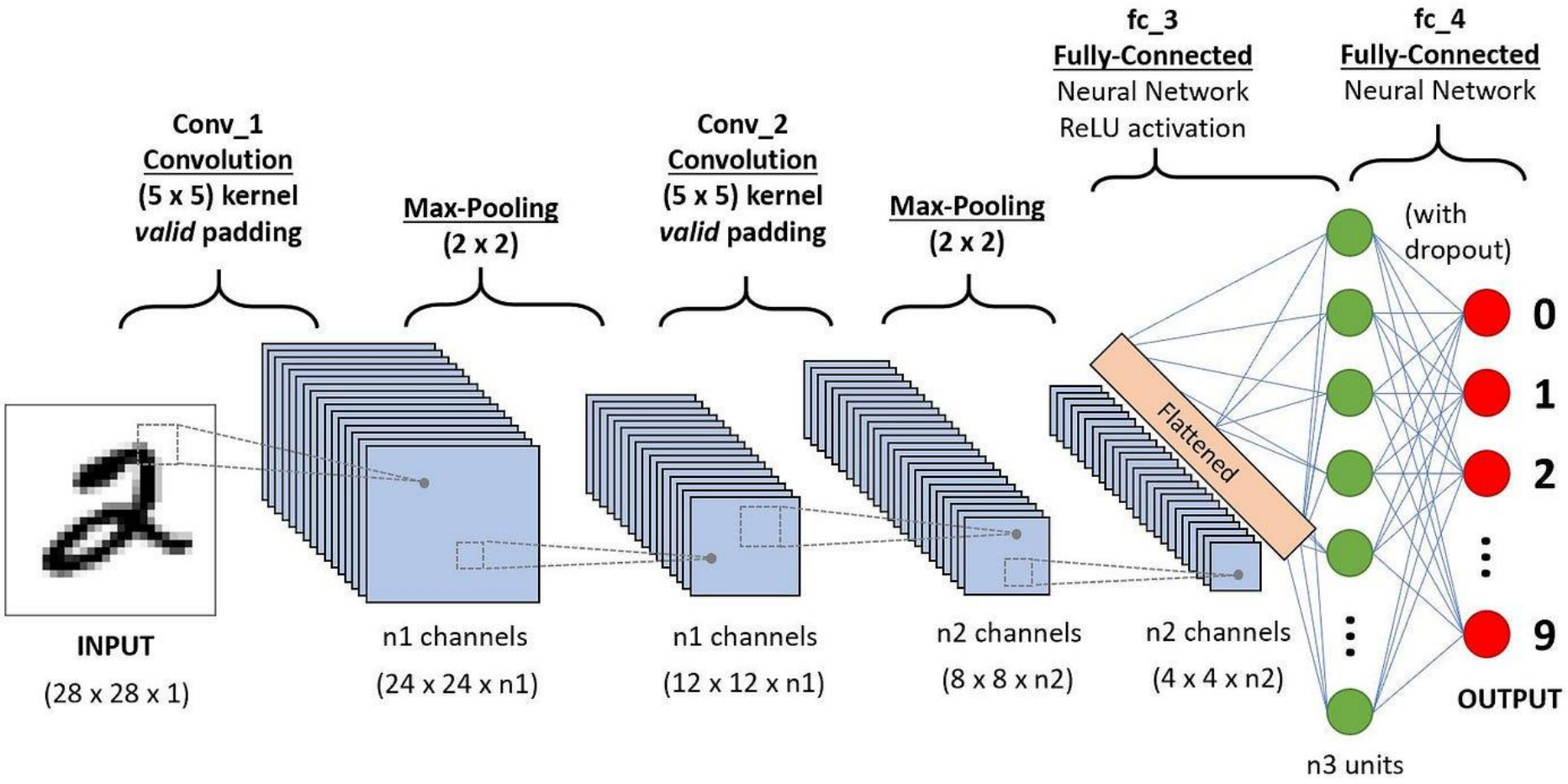
Transfer Learning (Optional)



Typically, training a model takes a large amount of compute resources and time.
Using a pre-trained model as a starting point helps cut down on both.

2-2. Image Classification Pipeline

Training Convolutional Neural Network

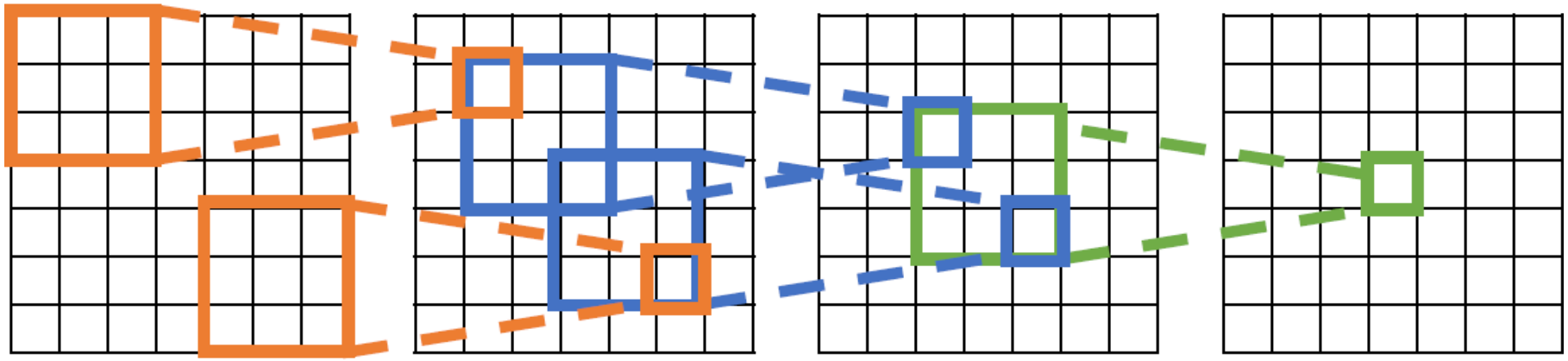


CNN consists of **convolution layer**, **pooling layer**, and **fully connected layer**

2-3. Characteristic of CNN

Receptive Field

Each successive convolution adds K – 1 to the receptive field size.
With L layers the receptive field size is 1 + L * (K – 1).



Problem: For large images we need many layers for each output to "see" the whole image image

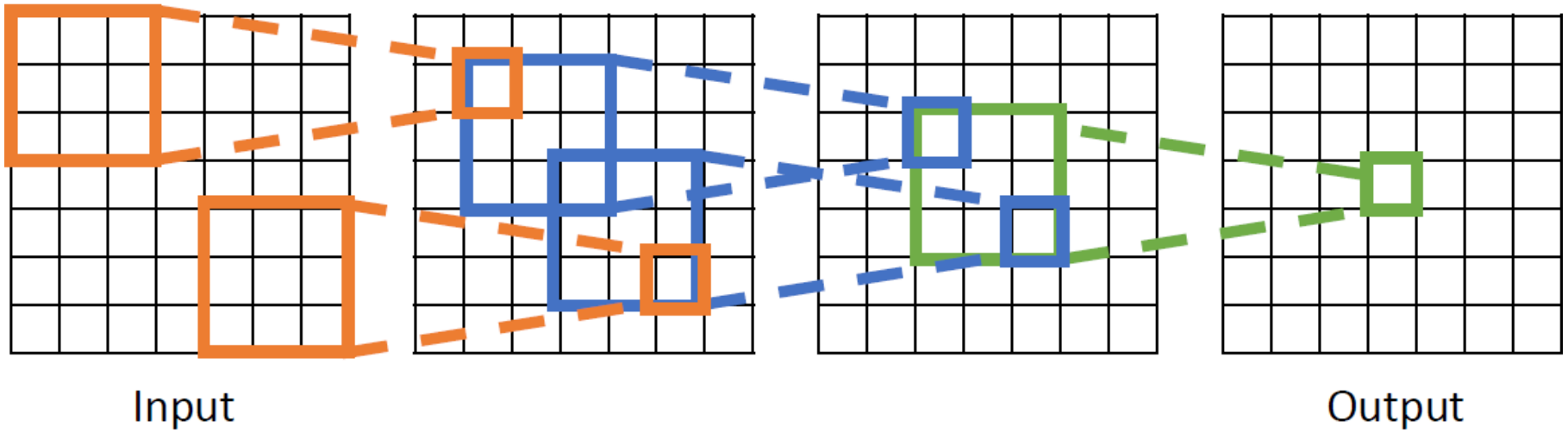
Solution: Downsample inside the network

2-3. Characteristic of CNN

Receptive Field

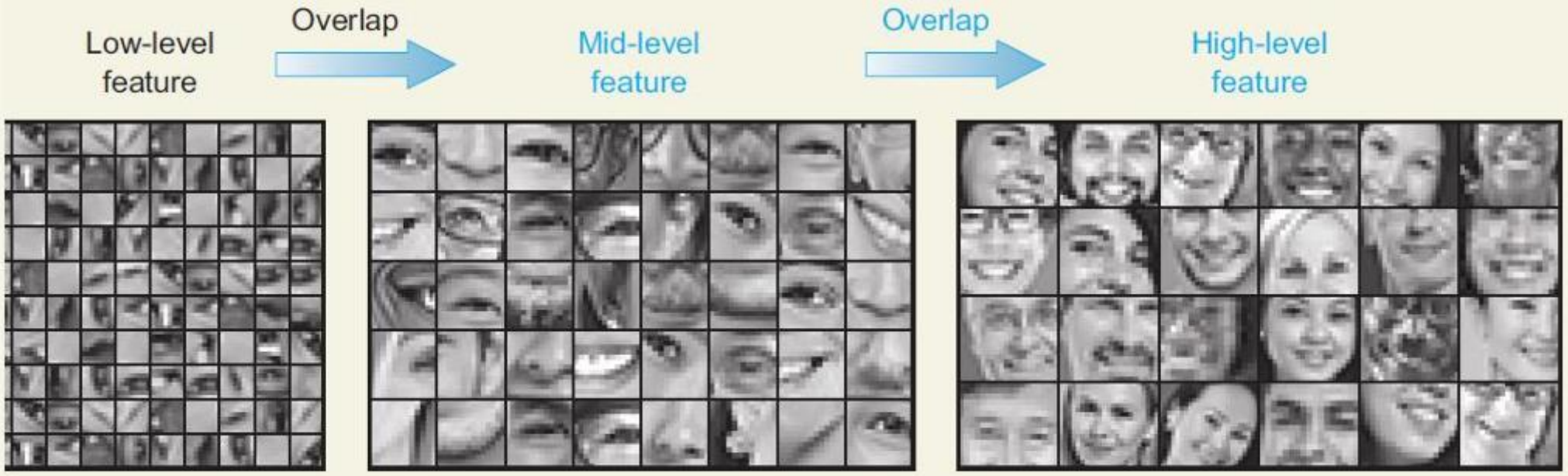
Q : 5 X 5 kernel vs 3 X 3 kernel

1. Receptive Field
2. Parameter
3. Non-linearity



2-3. Characteristic of CNN

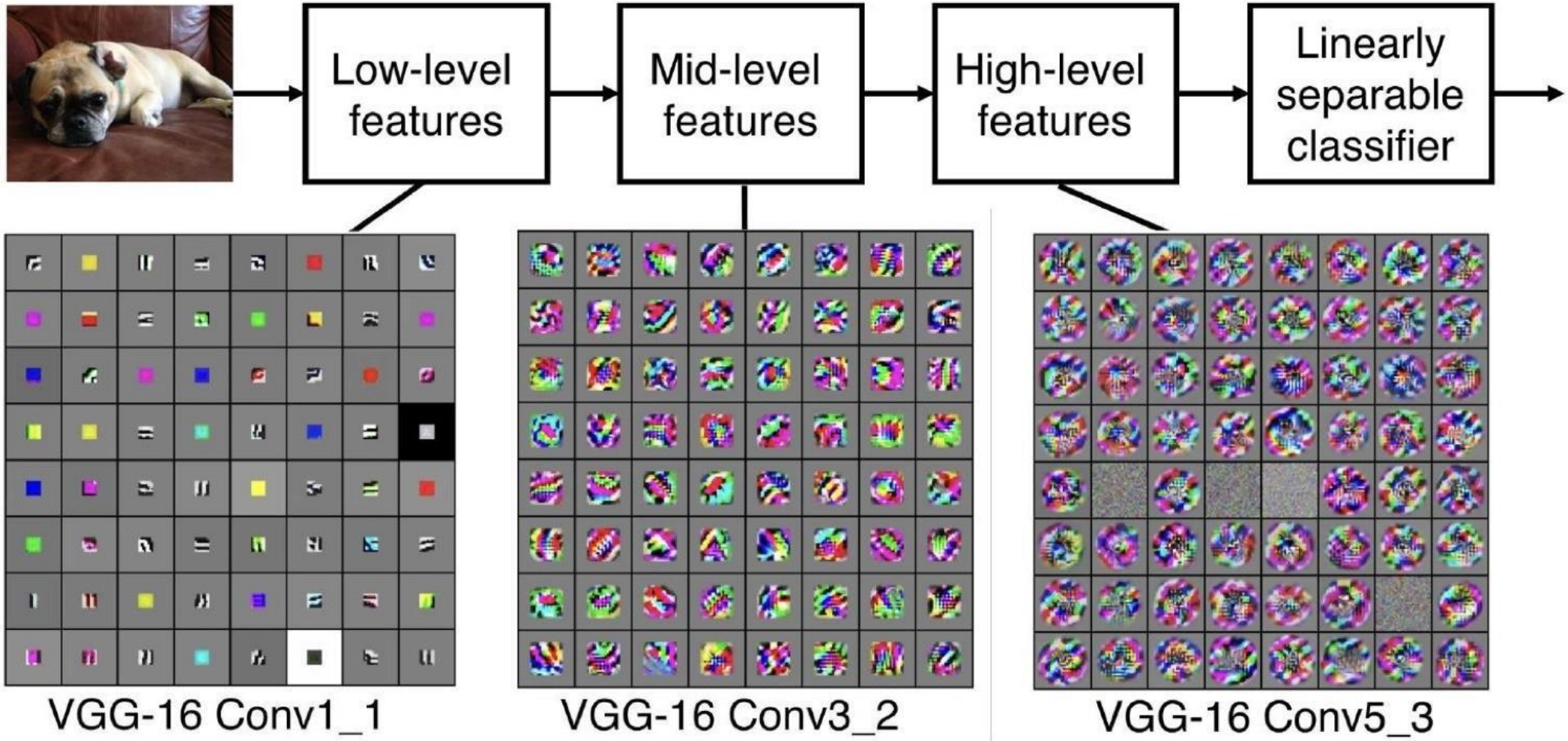
Feature Map



A simplified version of how CNNs learn faces

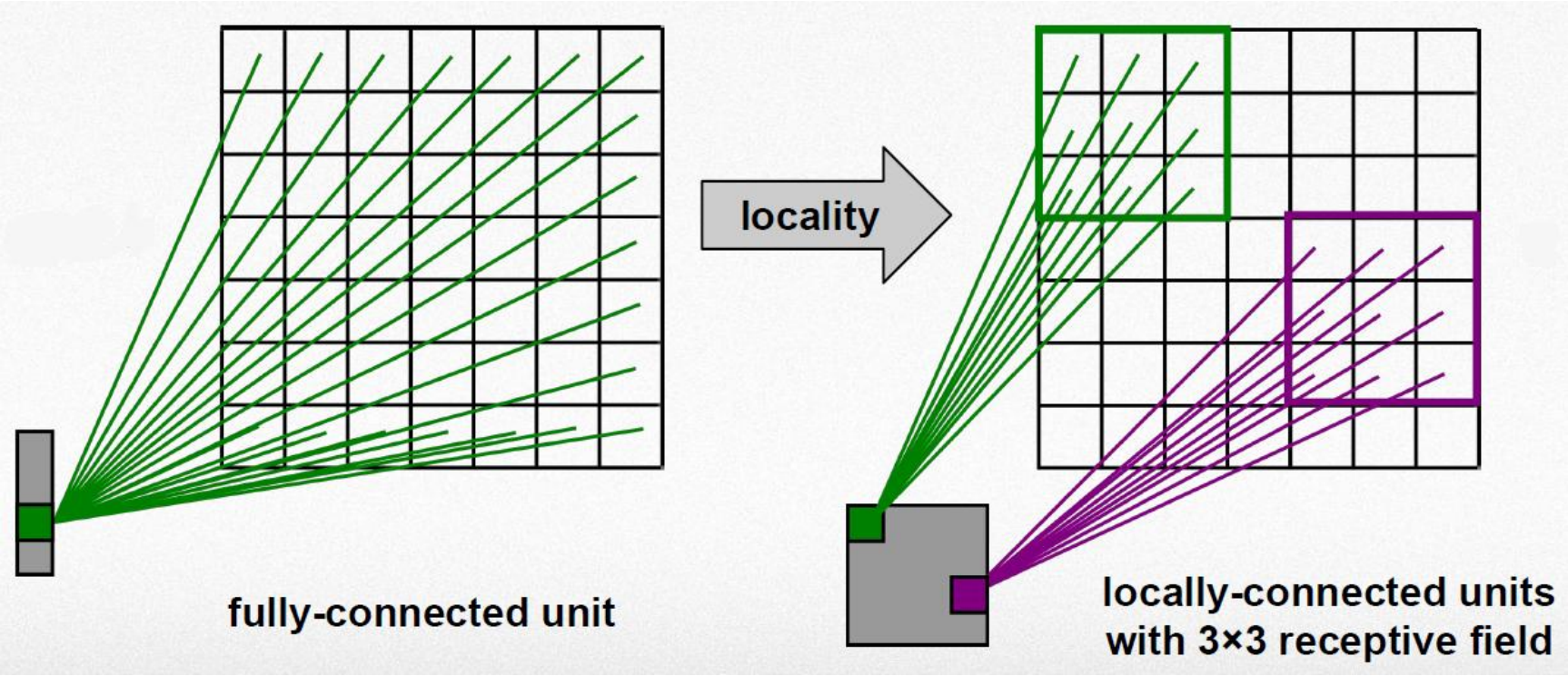
2-3. Characteristic of CNN

Feature Map



2-3. Characteristic of CNN

Locality



2-3. Characteristic of CNN

Invariance

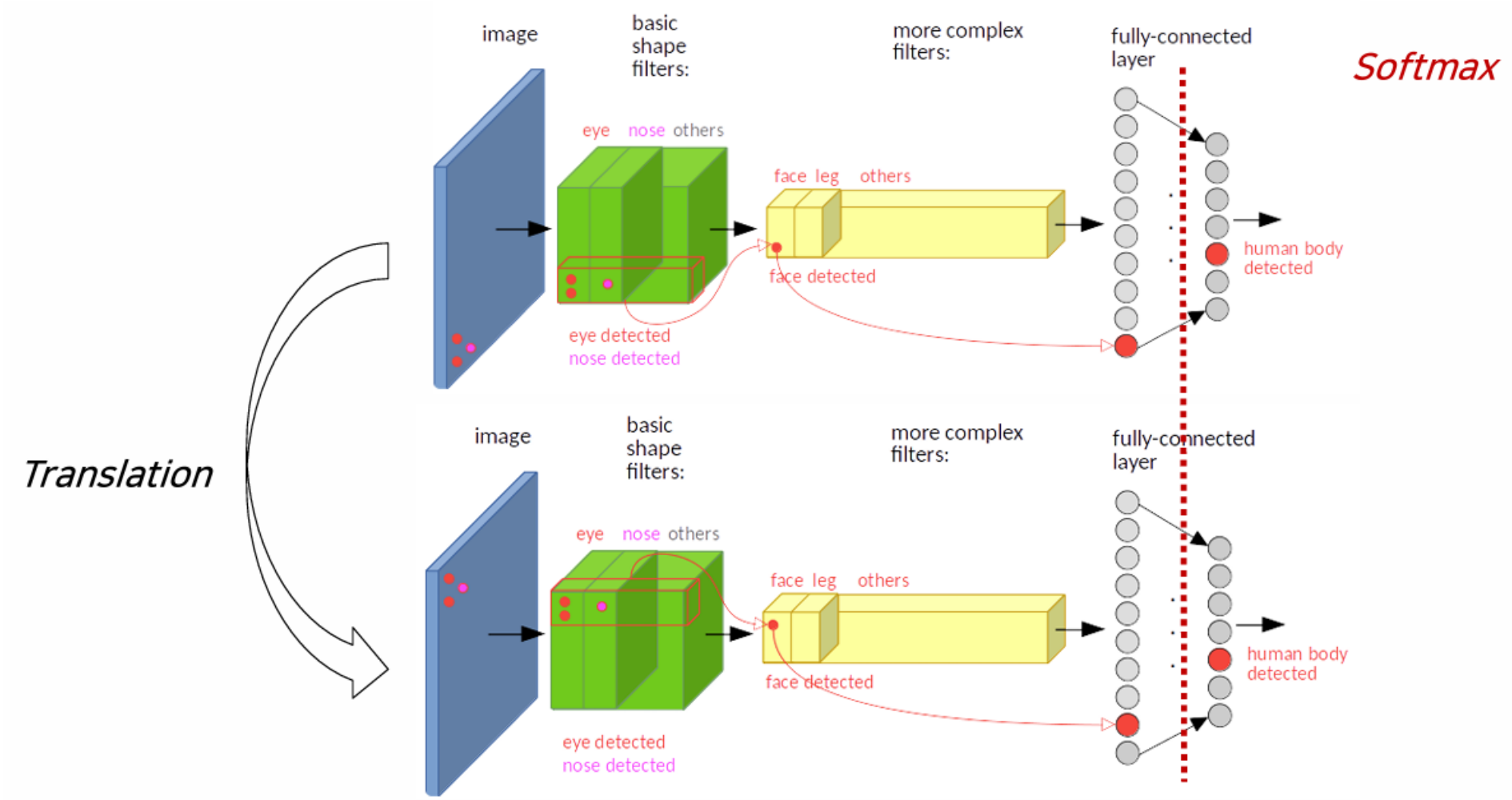
Cat

Cat

output remains consistent or is insensitive to translations (shifts) in the input

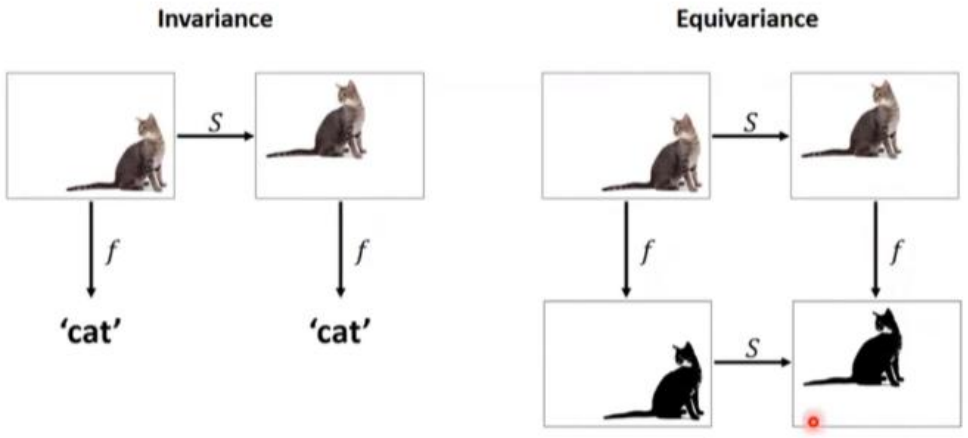
2-3. Characteristic of CNN

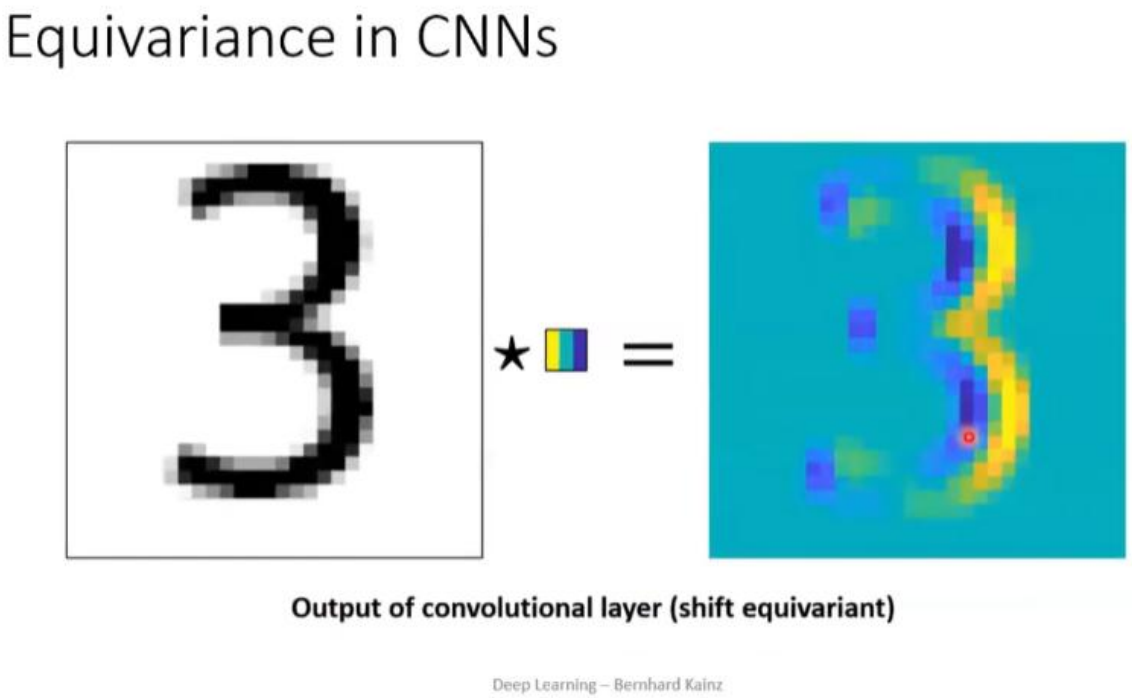
Invariance



2-3. Characteristic of CNN

Equivariance





input causes a corresponding translation (shift) in the output